

Probabilistic Prediction of Local Laser-Track Width from Thermal Imaging in Directed Energy Deposition

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Executive Summary

The study examines whether thermal camera measurements taken while the laser is melting metal can predict the appearance of the final solidified track. We tested four tracks made at laser powers of 200, 300, 350, and 400 W and compared their thermal behavior during manufacturing with profilometer measurements taken afterward. Because the profilometer didn't directly provide track width, and much of its data was missing, we developed our own method to extract the width from the available data. We estimate that these width measurements are accurate to about ± 0.05 mm.

We tested whether thermal measurements could predict final track geometry at laser powers the model had never seen before. Across eight model families, Ridge regression performed best, achieving errors of 0.085-0.195mm (mean 0.117mm), about 2-4 times better than the baseline, with simpler models outperforming more flexible ones. However, five independent tests showed that while thermal data could predict overall track width, it could not reliably predict the small fluctuations in width along each track. We also found that surface roughness increased with laser power, ranging from 1.26 μm at 200 W to 3.41 μm at 400 W. Finally, we identified and removed two sources of data leakage, increasing confidence that the reported model performance reflects true prediction rather than information accidentally shared between training and testing data.

We also tested how reliably the models could communicate uncertainty at unseen laser powers. Standard Gaussian-process intervals claimed 90% confidence but contained the true value only 24-63% of the time, showing that the model was more confident than it should have been when extrapolating to new powers. Conformal intervals improved this coverage to 76%, but uncertainty remained when operating outside the training conditions. Together, these results define a practical boundary for thermal monitoring: predict track geometry when the model has sufficient evidence, provide calibrated uncertainty estimates for those predictions, and raise an alert when conditions fall outside the model's reliable range.

1. Problem Formulation

In laser metal manufacturing, a focused laser creates a melt pool whose behavior affects the final shape and quality of the part. Because defects may only become visible after production, sometimes requiring destructive testing, there is a need for methods that can predict quality during manufacturing. This study tests whether thermal images captured during the process can predict the final local width of a laser track. We analyzed four single tracks produced at 200, 300, 350, and 400 W. Each was recorded with a thermal camera during manufacturing and later measured with an optical profilometer. We aligned the thermal and width data over the 20-100 mm region, enabling thermal features at each position to predict the final width at that location.

With only four tracks, one per power, we used leave-one-track-out validation to test whether models trained on three powers could predict a track at an unseen power. We evaluated prediction accuracy, the ability to capture small width changes along the track rather than only its average width, and the reliability of the model's confidence intervals.

2. Methodology

2.1 Data. Each track contains two measurements from the same physical location: thermal images of the melt pool during manufacturing and optical profilometer measurements of the final surface. This allowed us to match thermal behavior with the resulting geometry at each position. Frame counts and missing-data levels matched the provider's reference values, confirming dataset integrity.

2.2 Target construction. The profilometer measures surface height rather than track width, and 37-55% of its pixels are missing, while the plate's curvature makes simple height thresholding unreliable. Instead, we used the pattern of valid measurements to identify the track: the smooth, solidified track is measured more consistently, while the rough surrounding surface produces more missing data. This allowed us to identify the track boundaries and extract width, which we averaged into 0.2 mm bins to align with the thermal data. Although the challenge's 20-100 mm window removes the main laser startup effects, width extraction became less reliable near the track ends due to increased gaps in the profilometer data. We therefore used the reliable 25-97 mm region for training and evaluation, leaving 1,422 of 1,600 bins for modeling. Testing different extraction thresholds changed the mean widths by no more than ± 0.05 mm, which we used as the minimum uncertainty of the width measurements. Plate curvature was removed before calculating height-based features.

2.3 Feature construction. We derived fifteen thermal features across five physics-based categories: melt-pool geometry, intensity, stability, drift, and cooling behavior. A saturation check confirmed that peak intensity was not limited by the sensor and that the sensor could be used reliably. Before modeling, we removed two features: one that showed almost no variation across the data (saturation fraction) and another that leaked information about the final width (dropout density), thereby preventing misleading model performance.

2.4 Models, validation, and evaluation. We compared eight regression models, ranging from simple linear methods to more flexible neural and probabilistic approaches. Because neighboring slices are very similar, random train-test splits could allow the models to memorize patterns rather than predict new conditions. Instead, we used leave-one-track-out validation, training on three tracks and testing on the fourth, whose laser power was unseen during training. The lowest- and highest-power tracks also tested the harder case of predicting beyond the training power range. We evaluated models using MAE for prediction error, R^2 for overall fit, shape- R^2 for local width changes, and interval coverage for the reliability of uncertainty estimates.

2.5 Use of generative AI. Claude (Anthropic) assisted with planning, domain explanation, pipeline code generation, results auditing (e.g., sentinel-value detection, track-power verification), and drafting. ChatGPT (OpenAI) helped with grammar and simplification. The team reviewed all outputs, ensuring they were validated against known physics (monotonic power-width) and audited for leakage. All analytical decisions and interpretations belong to the team.

3. Modeling and Outcomes

3.1 Track width is predictable at unseen powers. Table 1 summarizes performance on unseen laser powers. Ridge regression performed best, with a mean error of 0.117 mm and its strongest advantage at the two extreme powers: 0.195 mm at 400 W and 0.085 mm at 200 W, compared with baseline errors of 0.394 and 0.359 mm—a two-to-four-fold improvement where prediction was most difficult. Lasso (0.118 mm) and the linear Gaussian process (0.118 mm) produced similar results, while nonlinear models struggled to predict beyond the powers seen during training. This consistent performance across multiple models shows that the simpler linear relationship was supported by the data rather than assumed. Figure 1 compares the measured and predicted widths for the held-out 200 W track, with predictions remaining largely within the measurement uncertainty band.

Table 1. Leave-one-track-out mean absolute error (mm). Tracks 8 and 21 require extrapolation beyond the training power range.

Model	Track 8	Track 10	Track 14	Track 21	Mean
Linear					
Baseline (mean)	0.394	0.099	0.085	0.359	0.234
Ridge (ours)	0.195	0.114	0.074	0.085	0.117
Lasso	0.192	0.116	0.072	0.090	0.118
GP (linear kernel)	0.199	0.114	0.077	0.081	0.118
Non-Linear					
GP (RBF kernel)	0.214	0.127	0.101	0.283	0.181
SVR (RBF)	0.322	0.147	0.122	0.305	0.224
k-NN (k=25)	0.287	0.258	0.113	0.218	0.219
Gradient boosting	0.326	0.240	0.096	0.230	0.223
MLP (32,16)	0.216	0.111	0.149	0.622	0.274

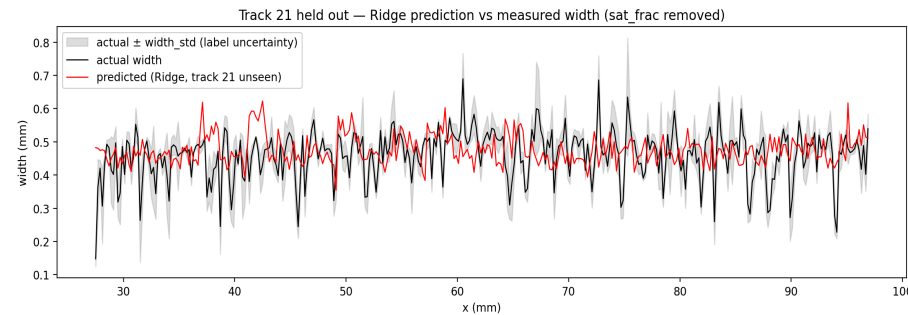


Figure 1. Measured vs. predicted width along held-out Track 21 (200 W — extrapolation below the training power range). Gray band: $\pm 1\sigma$ width-extraction-truth label uncertainty.

3.2 Slice-scale variation is unpredictable — a verified negative result. Mean-subtracted scoring yields shape- $R^2 \leq 0$ for every model on every track, indicating that no model outperforms a simple flat line at each track's mean. This null result persisted through five independent checks — leakage removal, label smoothing, alignment scans (no consistent lag), within-track feature-width correlations ($|r| \leq 0.25$ throughout), and a position-prediction test showing thermal features cannot infer scan position. The residual width jitter is likely measurement-scale noise (width-extraction-truth floor ± 0.05 mm), not a missed signal. Thermal imaging can predict the track level; however, slice-to-slice fluctuations around that level are not recoverable at 0.2 mm resolution.

3.3 Track variation descriptors. Surface roughness increased steadily with laser power, from 1.26 μm at 200 W to 3.41 μm at 400 W, while waviness ranged from 4.5-5.8 μm across the four tracks. Contrary to our initial hypothesis, greater thermal instability did not correspond to greater surface roughness: Tracks 14 and 21 were about three times more thermally unstable than Tracks 8 and 10, yet were among the smoothest. Within each track, thermal flicker and roughness were also nearly uncorrelated ($|r| \leq 0.05$). These results suggest that roughness is more closely associated with laser power than thermal instability in this dataset, although conclusions are limited by the four-track sample.

3.4 Uncertainty under extrapolation. Uncertainty estimates were less reliable at unseen laser powers. Although the linear Gaussian process reported 90% confidence intervals, they contained the true width only 24% of the time on Track 8 and 63% of the time on Track 10, indicating that the model was overconfident when conditions shifted beyond its training experience. Conformal intervals improved coverage to 76% at the 90% confidence level but could not fully correct for extrapolation. Based on this limitation, we propose a two-part monitoring approach: use conformal intervals to monitor predictable track width and an anomaly alarm to detect unusual thermal behavior. In a case study of two known collapse events, a more sensitive alarm detected both events, raising 65 total flags, while a stricter setting detected only one event, raising 32 flags. Because only two known events were available, these results demonstrate feasibility rather than a validated detection rate.

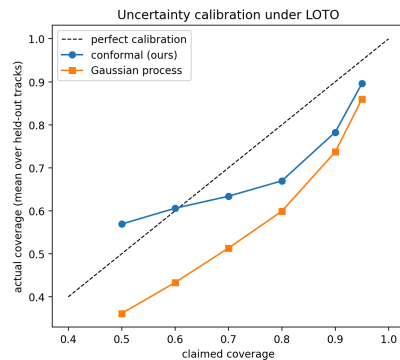


Figure 2. Claimed vs. achieved interval coverage on held-out tracks. Both methods fall short of perfect calibration; conformal improves coverage but does not fully restore it.

3.5 Limitations and uncertainties. This study is limited to four tracks and four laser powers, so the findings, especially the strong performance of linear models, should not be generalized beyond the tested power range. Melt depth could not be measured without destructive testing, and all local-width results are limited to the 0.2 mm resolution and ± 0.05 mm measurement uncertainty of this study. Linear models may perform well because the thermal images were already converted into physics-based features, capturing much of the nonlinear behavior before modeling. With only four power levels, more complex relationships are also difficult to identify, and no nonlinear model outperformed the linear approaches even when predicting within the tested power range.

4. Conclusion

Thermal monitoring can predict overall track geometry at unseen laser powers, with width errors of 0.085-0.195 mm. Pool size and intensity were informative for overall width, whereas small slice-to-slice fluctuations were not reliably predictable, as confirmed by five independent checks. Thermal instability also increased sharply at lower power levels, providing an additional measure of process stability for future study.

These findings are especially relevant to applications such as patient-specific implants and aerospace components, where destructive inspection can be costly or impractical. In situ monitoring with calibrated uncertainty estimates could help assess quality during manufacturing and flag abnormal regions for targeted inspection. Our framework provides a small-scale foundation for this approach: predict what can be predicted reliably, communicate uncertainty honestly, flag what cannot, and audit the modeling pipeline for data leakage.

References. [1] NSF Future Manufacturing Data Challenge dataset and starter loading code, Texas A&M Institute of Data Science, Zenodo record 21285367, 2026.